

Proof. Let

$$\mathbf{a} = (a_1, \dots, a_n), \quad \mathbf{b} = (b_1, \dots, b_n) \in \mathbb{R}^n,$$

and denote the standard Euclidean inner product and norm by

$$\langle \mathbf{a}, \mathbf{b} \rangle = \sum_{i=1}^n a_i b_i, \quad \|\mathbf{a}\| = \sqrt{\langle \mathbf{a}, \mathbf{a} \rangle} = \sqrt{\sum_{i=1}^n a_i^2}.$$

For any real scalar t consider the non-negative quadratic form

$$f(t) = \|\mathbf{a} + t\mathbf{b}\|^2 = \sum_{i=1}^n (a_i + tb_i)^2 \geq 0 \quad (t \in \mathbb{R}).$$

Expanding the square and collecting terms gives

$$\begin{aligned} f(t) &= \sum_{i=1}^n (a_i^2 + 2t a_i b_i + t^2 b_i^2) \\ &= \underbrace{\sum_{i=1}^n a_i^2}_A + 2t \underbrace{\sum_{i=1}^n a_i b_i}_B + t^2 \underbrace{\sum_{i=1}^n b_i^2}_C \\ &= A + 2Bt + Ct^2. \end{aligned}$$

Thus $q(t) := Ct^2 + 2Bt + A$ satisfies $q(t) = f(t) \geq 0$ for every $t \in \mathbb{R}$.

Case $C > 0$. Since the leading coefficient C is positive, a quadratic is non-negative on $\mathbb{R}$ iff its discriminant is non-positive:

$$(2B)^2 - 4AC \leq 0.$$

Dividing by $4 > 0$ yields

$$B^2 \leq AC.$$

Case $C = 0$. Then $\mathbf{b} = \mathbf{0}$, whence $B = 0$ and the desired inequality reduces to $0 \leq 0$, which holds.

Combining the two cases we obtain

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right),$$

i.e. the Cauchy-Schwarz inequality.

Equality case. Equality in the discriminant condition occurs exactly when

$$(2B)^2 - 4AC = 0 \iff B^2 = AC.$$

If $C > 0$, this means that $q(t)$ has a double root $t_0 = -\frac{B}{C}$, and therefore

$$\mathbf{a} + t_0 \mathbf{b} = \mathbf{0} \implies \mathbf{a} = -t_0 \mathbf{b}.$$

Thus the vectors are linearly dependent: there exists $\lambda \in \mathbb{R}$ such that $\mathbf{a} = \lambda \mathbf{b}$, i.e. $a_i = \lambda b_i$ for all i . Conversely, if $\mathbf{a} = \lambda \mathbf{b}$ for some λ , then both sides of the inequality are equal. Hence equality holds iff the two vectors are proportional. $\square$